

Inha University in Tashkent

Theme 9. Epistemology: Scientific Knowledge and Its Forms

Tashkent – 2024

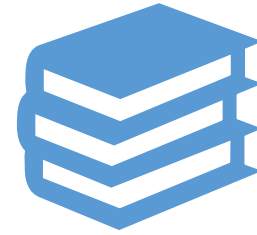
Outline:



Etymology of Epistemology
and Gnosiology



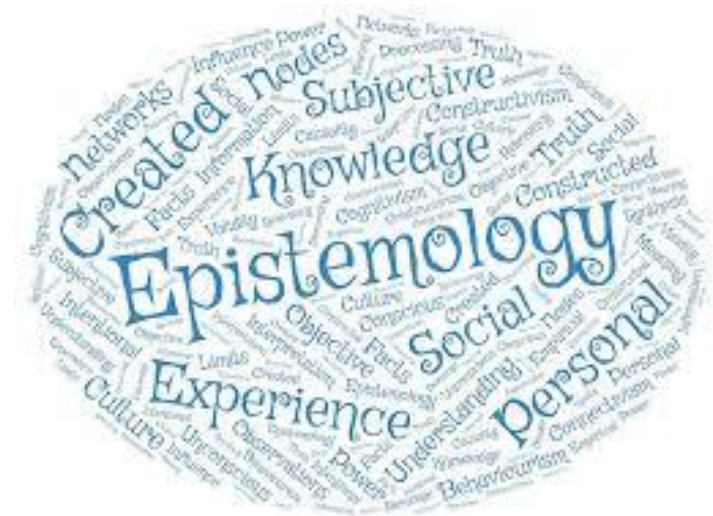
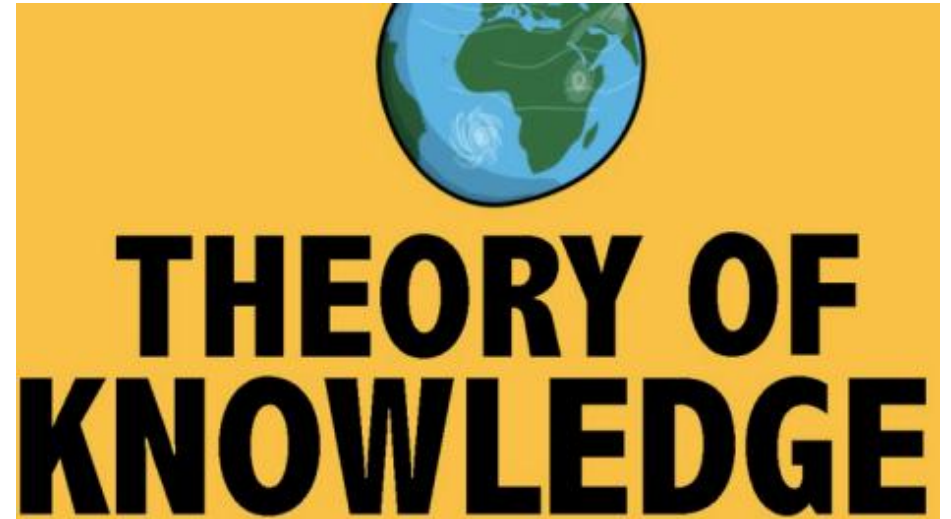
Philosophical Trends on
Epistemology



The Forms of Knowledge

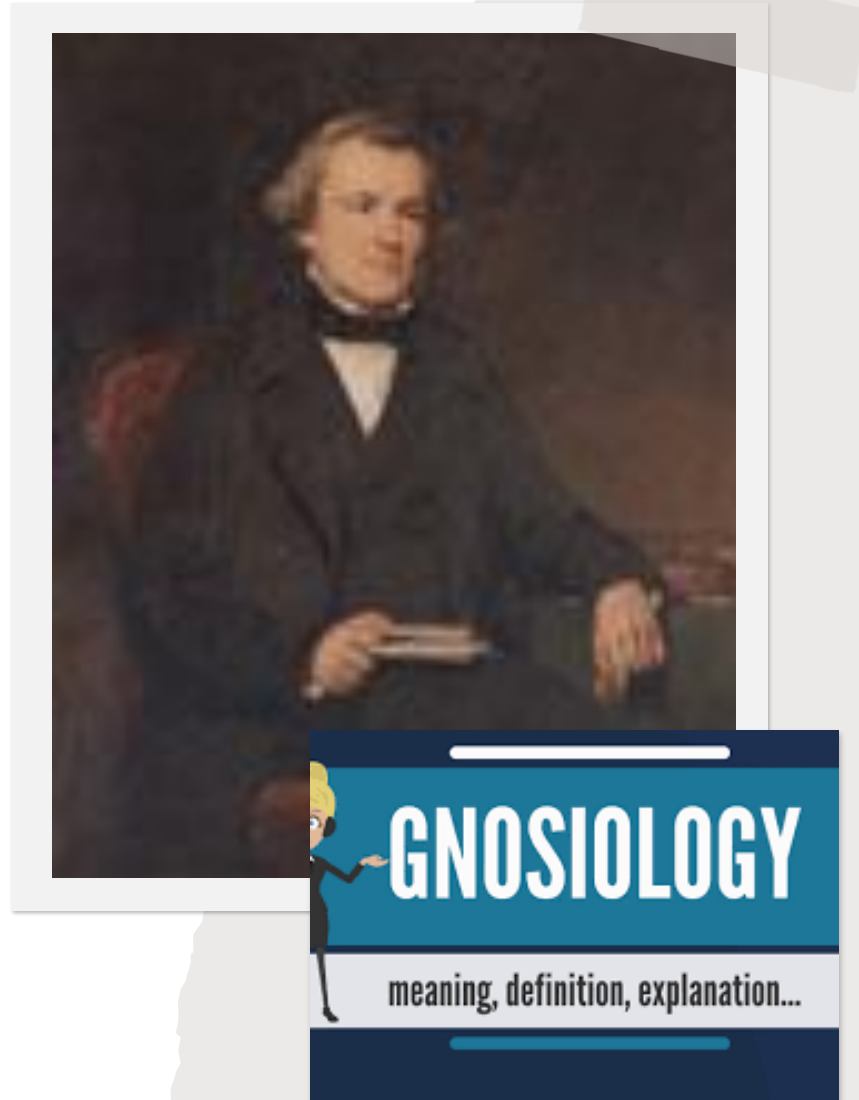
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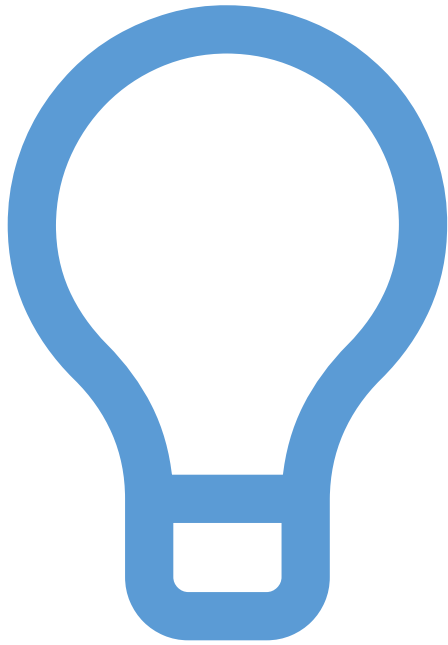
- Gnosiology and epistemology have their distinguishing feature among the disciplines that aimed to examine the human being, the knowledge of man and the process of cognition.
- Gnosiology studies about the essence of knowledge and its possibilities, the attitude of human being towards the existence, the way which human knowledge is perceived and the means of determining the authenticity of knowledge.
- The term was firstly used by German philosopher Alexander Baumgarten (1714-1762) who developed the aesthetic philosophy (a branch of philosophy that explores the nature of art, beauty and taste, with the creation and appreciation of beauty).
- If gnosiology is the concept on general knowledge, epistemology is basically cognition that based on scientific activity. Therefore, these terms had different meanings and usage. Nevertheless, they constitute the same implication in some western philosophical literature of modern period.



Features of Gnosiology and Epistemology

- The Scottish philosopher James Frederick Ferrier (1808–1864) was firstly introduced the term “epistemology”.
- He designated it as a branch of philosophy which aims to discover the meaning of knowledge. Later, French philosophers gave the term a narrower meaning as “theory of knowledge”.
- As its characteristic aspect epistemology basically deals with scientific analysis that is the source for scientific knowledge whereas there are several types of general knowledge such as ideological, artistic, religious cognition.
- Some scientists argue that epistemology as a scientific knowledge is a special section of gnosiology that has its own objectives and problems.
- Ontology examines the general doctrine and the fundamental principles of existence which are the main source for gnosiology.





The Questions of Knowledge

- The scientists and philosophers posed some questions on the concept of cognition. These are followings:
 - 1) *Do we know the world?*
 - 2) *What are the boundaries of knowledge?*
 - 3) *How do we know that we know?*
- The answers to the first question are different in accordance with the philosophical schools.
- The followers of certain ideas were classified into the particular philosophical trends.

Philosophical Trends on Epistemology

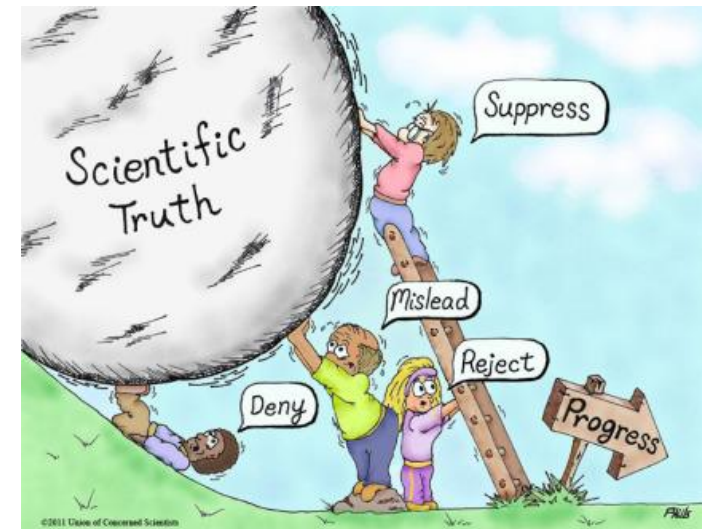
- *Gnoseological optimism* – the world is cognized, there is no knowledge boundary, only time and means are needed.
- *Agnosticism* – the world is unknowable in principle; a person does not know the world but builds a virtual world on the basis of sensory perception.
- *Skepticism* – we know the phenomenal (sensible) world, the knowing ability of the real world is problematic

AGNOSTICISM

Always Doubt
Always Question
Challenges you to think with
Logic and Reason

ATHEISM

No Doubt
No Question
Challenges you to disprove their
Lack of Belief

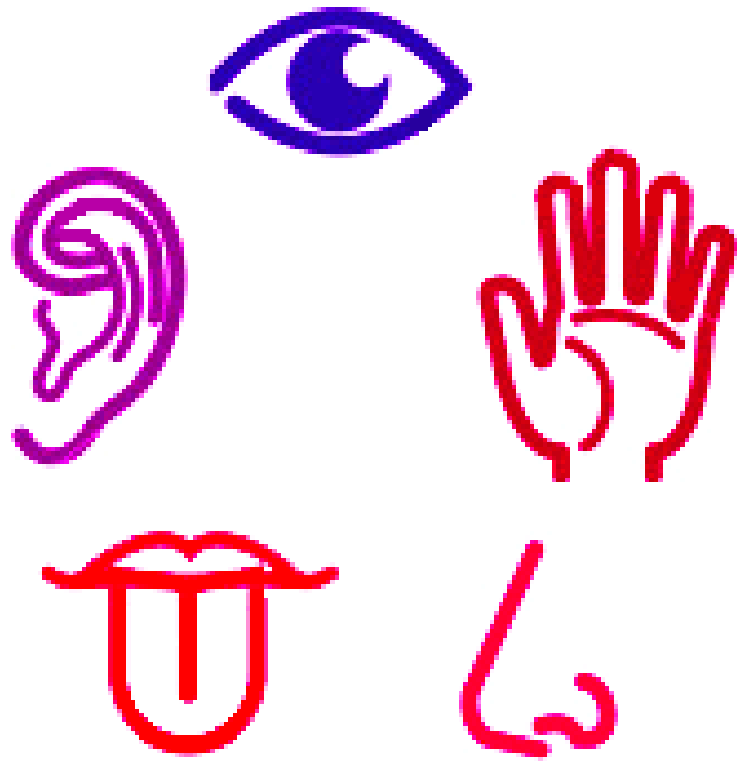




Forms of the cognition

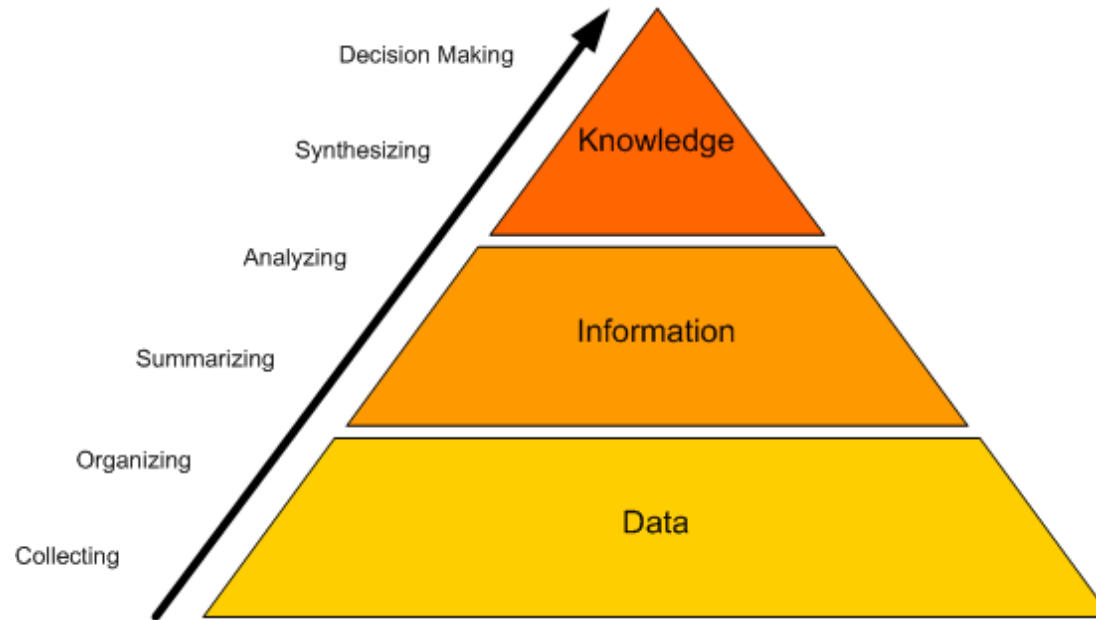
- ***Sensual cognition*** – is the level of sensations, perceptions and notions.
 - ***Rational cognition*** – is the level of abstractions expressed in hypotheses, theories, laws, and cause-effect relationships. At the level of rational cognition, a person is able to construct a model of an event so that its action is most effective. Forms of rational knowledge: concept or idea, judgment and reasoning.
 - ***Supersensible knowledge*** – is intellectual intuition, metaphysics, direct knowledge, drawn by the subject from the depth of itself. This kind of knowledge is especially common in the mystical currents of traditional religions.
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Sensual cognition



- The base of human knowledge is first of all related to the emotion.
- Commonly recognized sensory organs (or systems) are those: vision, hearing, touch, taste, smell. Sense systems are the receivers from the physical world to the realm of the mind where we interpret the information, creating our perception of the world around us.
- There are three forms of emotional cognition: intuition, perception, imagination.
- Observation is the starting point for experimental knowledge. It is primarily a targeted study of sensory systems based on human emotion, perception, imagination, during which we learn about the external aspects of the object, its features and signs.

Rational cognition



- Human thinking is a complex process of learning, covering many methods, styles and forms of knowledge.
- According to the classical formal logic, there are forms of thinking (judgment) and conclusions (consequence, entailment).
- Judgment is the form of the idea, by which the conformity of the concepts is confirmed or denied.
- The experiment is a specific target method of studying phenomena that can be recreated and controlled by the researcher exactly how this happens.
- The experimental style arose on the basis of physics, and then widely used in chemistry, biology, physiology and other natural sciences.

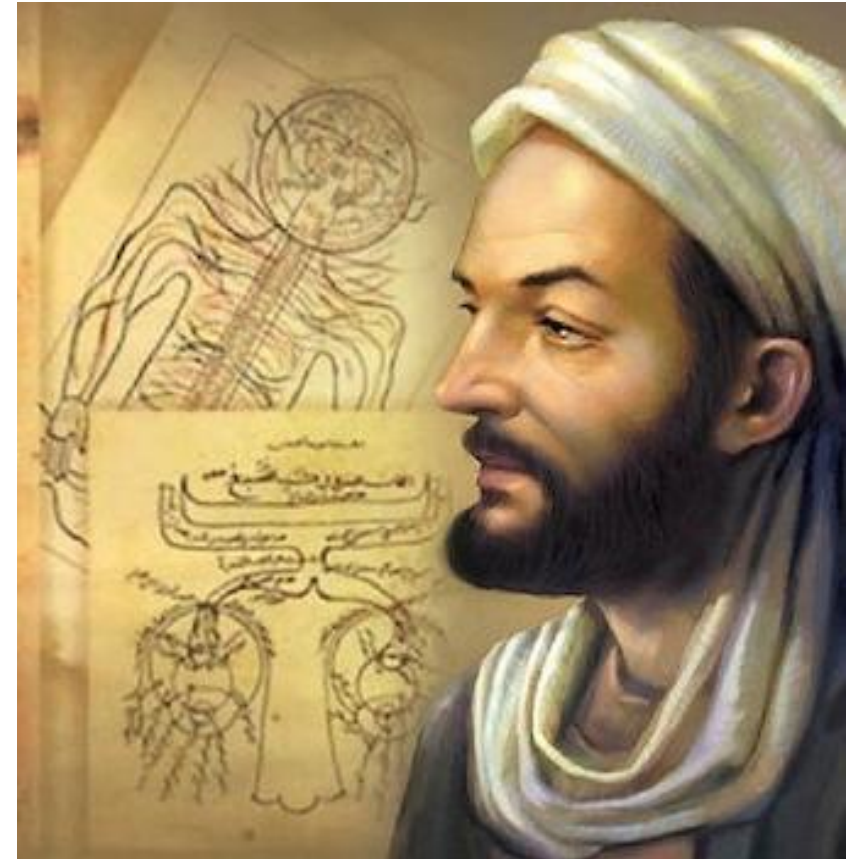
Abu Rayhan Beruni



- Some works of Abu Rayhan Beruni are devoted to the issue of cognition.
- There are two reasons for interest in cognition. First of all, the process of cognition is a special flavor. Secondly, the purpose of knowledge is to satisfy the needs of the people.
- The process of cognition begins with the information that provided by sense organs. They serve as a kind of auxiliary and foundation for mental cognition.
- The sincerity of knowledge is determined through observation and experimentation.

Abu Ali ibn Sina

- Abu Ali ibn Sina (980-1037) is a great scholar and thinker. He was born in the village of Afshona, near Bukhara and died in Hamadon.
- The scientific works of Ibn Sina are more than 280 names. They are dedicated to medicine, philosophy, logic, psychology, ethics, music, pharmacology and other areas.
- Scientist's works "The Book of Healing", "The Canon of Medicine", "The Book of Knowledge" and "Nature" are well-known. Ibn Sina followed Farabi on the interpretation of the universe. In particular, "*Wujudu wājib*" (God) is the first reason and "*wujudi mumkin*" (all material and spiritual beings) is created by "*Wujudu wājib*" – God.



Abu Ali ibn Sina



- Ibn Sina's major work on "The Book of Knowledge" examines reinvention of epistemology.
- It explains his theory of knowledge in which intentionality acts as an interaction between the mind and the world. This, in turn, led Ibn Sina to distinguish an operation of intentionality specific to the generation of numbers.
- The starting point of Avicenna's logic is that all knowledge is either forming concepts (*taṣawwur*) by means of definitions or acknowledging the truth (*taṣdīq*) of a categorical statement by means of syllogisms.

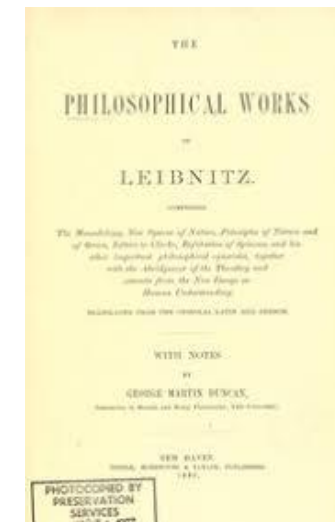
John Locke

- John Locke (1632-1704) was English philosopher and outstanding representative of British school of philosophy whose works lie at the foundation of modern philosophical empiricism and political liberalism. He was considered as an inspirer of both the European Enlightenment and the Constitution of the United States. His philosophical thought was close to that of the founders of modern science, especially Robert Boyle, Sir Isaac Newton, and other members of the Royal Society of London.
- John Locke was an explorer of British Empiricism and in turn, the empiricists hold the view that all human knowledge must come directly or indirectly from the experience of the world that we acquire through the use of our senses alone. This contrasts with the thinking of the rationalist philosophers, such as Rene Descartes, Benedictus Spinoza and Gottfried Leibniz who stated that in principle, at least, it is possible to acquire cognition solely through the use of reason. Locke sets out to demonstrate that the best explanation of the world as we experience it is corpuscular theory. Indeed, corpuscular theory was becoming popular in 17th century scientific thinking and is key to Locke's view of the physical world.



Gottfried Leibniz

- Gottfried Wilhelm Leibniz (1646-1716) was German philosopher and mathematician. Gottfried Leibniz is famous in mathematics for his invention of the so-called “infinitesimal calculus”. But both Leibniz and Newton claimed the invention as their own. It seems clear that they had in fact reached it independently, but Leibniz developed a much more usable notation, which is still used today.
- He was engaged in metaphysics and logic. Gottfried Leibniz was a follower of rationalism and his major work on the issue was “Monadology”. He asserts that in principle all can be accessed by rational reflection. However, due to shortcomings in our rational faculties, human beings must also rely on experience as a means of acquiring knowledge.



Thank you for your attention!